

Water Intake Calculator and Tracker in C++

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Abstract—Adequate daily hydration depends on body weight, climate, and physical activity, yet most people estimate their needs by guesswork. This paper presents a console-based Water Intake application written in C++ that computes a personalized daily water target, records each drink, tracks progress against the target, and maintains a daily streak counter. The program stores its state in a single struct and is organized into three functions: a target calculator, a drink logger, and a dashboard. The target is computed as 35 ml per kilogram of body weight plus a climate adjustment and an activity adjustment. We describe the design and the program’s inputs, present its output, and analyze its strengths and limitations.

Keywords—water intake, hydration, C++, struct, console application, streak tracking

I. INTRODUCTION

Water is essential for thermoregulation, nutrient transport, and normal organ function. Dietary reference values for water are expressed per day and vary between individuals [1], [2]. A fixed recommendation such as “eight glasses a day” ignores differences in body mass, ambient temperature, and activity level.

This project implements a simple but complete water-intake assistant in C++ [3]. The application asks the user for body weight, climate, and activity level, computes a daily target in milliliters, and then lets the user log each drink during the day. When the target is met, a streak counter increases to encourage consistent habits. The objectives of this report are (i) to describe the program design, (ii) to document its input and output, and (iii) to analyze its behavior and limitations.

II. SYSTEM DESIGN

A. Data Structure

All application state is held in one struct, `WaterIntakeApp`, which is passed by reference to each function. Table I lists its members.

TABLE I. MEMBERS OF THE `WaterIntakeApp` STRUCT

Member	Type	Purpose
weight	double	Body weight (kg)
isTropical	bool	Climate: true = tropical
activityChoice	int	1 light, 2 moderate, 3 heavy
dailyTargetMl	double	Computed daily target (ml)
currentProgressMl	double	Water logged today (ml)
streakCount	int	Consecutive days target met
reminderIntervalHours	int	Reminder period (default 2 h)

B. Functions

The program is divided into three functions, plus `main()`, which handles setup input:

- `calculateWaterTarget()`: computes the daily target from the user profile.
- `logWaterDrink()`: reads a drink amount, updates progress, and either reports the remaining volume or announces that the target is reached and increments the streak.
- `displayDashboard()`: prints the current status, taking the struct as a const reference so that it cannot modify the data.

C. Target Calculation

The daily target T in milliliters is given by

$$T = 35 \cdot W + C + A \quad (1)$$

where W is body weight in kilograms, C is the climate adjustment (+500 ml for tropical, +200 ml for subtropical), and A is the activity adjustment (0 ml for light, +350 ml for moderate, +700 ml for heavy activity). Table II summarizes the constants.

TABLE II. ADJUSTMENT FACTORS USED IN (1)

Factor	Condition	Added (ml)
Climate (C)	Tropical	500
Climate (C)	Subtropical	200
Activity (A)	Light	0
Activity (A)	Moderate	350
Activity (A)	Heavy	700

III. INPUT AND OUTPUT

A. Program Input

The program is interactive and reads all data from the keyboard through `cin`. Input occurs in two stages: a one-time setup in `main()`, followed by a menu loop. Table III lists every input.

TABLE III. PROGRAM INPUTS

Input	Type	Valid values	Use in program
weight	double	Weight in kg	Base water: $\text{weight} \times 35$
climateChoice	int	1 tropical, 2 subtropical	<code>isTropical = (choice == 1)</code>
activityChoice	int	1, 2, 3	Activity adjustment in (1)
menuChoice	int	1 to 4	Selects menu action
drinkAmount	double	Volume in ml	Added to daily progress

B. Setup Input

The setup screen, titled “Setup Water Intake Calculator,” first asks for body weight in kilograms. It then asks for the climate of the user’s residence (1. Tropis, 2. Subtropis) and for the daily activity level (1. Ringan, 2. Sedang, 3. Berat). The climate answer is converted to a boolean by `isTropical = (climateChoice == 1)`, so any value other than 1 is treated as subtropical. For activity, the program only tests for 2 and 3, so any other value is treated as light activity (no addition). After the three answers are read, `calculateWaterTarget()` computes the daily target using (1).

C. Menu Input

After setup, the program repeats a main menu until the user enters 4. Menu 1 displays the dashboard and streak tracker. Menu 2 calls `logWaterDrink()`, which asks for the amount of water drunk in milliliters. Menu 3 starts a new day: if the current progress is below the daily target, the streak is reset to 0, and in both cases the progress is reset to 0 ml. Menu 4 prints a thank-you message and exits. Any other value prints an “invalid menu choice” message and the menu is shown again.

D. Program Output

The program output was tested with the data of someone who is 60 kg, lives in the tropical climate, and a moderate activity level. Based on section 3, the formula given was used to calculate the daily water target which is 2950 mL.

After the setup process is completed, the program displays the main menu which allows the user to choose the options: the dashboard, record water intake, reset timer option, and exit the application.

```
=====
          SETUP WATER INTAKE CALCULATOR
=====
Masukkan Berat Badan (kg): 60

Pilih Iklim Tempat Tinggal:
1. Tropis
2. Subtropis
Pilihan (1/2): 1

Pilih Tingkat Aktivitas Harian:
1. Ringan
2. Sedang
3. Berat
Pilihan (1-3): 2
```

Fig. 1. Setup input and calculated daily target

For the first test, the user inputs 500 mL into the daily water intake calculator through the main menu which the program will display the remaining daily water intake needed and a reminder to drink in a constant intervals of hours. Along with the reminder of daily water intake, the program also displays the estimated amount of glass needed to fulfill the daily water intake, in which the program assumes that each glass taken is 250 mL of water.

```
=====
          LOG WATER INTAKE / DRINK
=====
Masukkan jumlah air yang diminum (ml): 500
[+] Berhasil mencatat 500 ml air.

=====

[NOTIFIKASI REMINDER]
Ingat untuk minum air lagi 2 jam dari sekarang!
Sisa target minum hari ini: 2450 ml (~9 gelas)
=====
```

Fig. 2. Logging a drink and remaining volume

For the second test, to fulfill the remaining daily water intake, the user inputs 2450 mL of water into the daily water intake calculator. Because the condition “`app.currentProgressMI >= app.dailyTargetMI`” is satisfied, the program will display that the daily water intake has been achieved and the user’s daily water intake streak counter increases.

```
=====
          LOG WATER INTAKE / DRINK
=====
Masukkan jumlah air yang diminum (ml): 2450
[+] Berhasil mencatat 2450 ml air.

=====

[!] SELAMAT! Target harian Anda tercapai!

=====
Streak harian Anda bertambah menjadi: 1 Hari 🍷
=====
```

Fig. 3. Target achieved message and streak update

Then once the daily water intake goal has been achieved, the user can go to the dashboard through menu 1. In the dashboard menu, the user is able to view the season status, daily water intake goal, the current daily water intake, the daily progress, and the water streak. After all the tests, it shows in the dashboard that the daily progress has reached 100% and the water streak increased by 1.

```
=====
          WATER INTAKE DASHBOARD
=====
Status Iklim       : Tropis (true)
Target Harian      : 2950.0 ml (3.0 Liter)
Terkumpul Saat Ini: 2950.0 ml
Progres Harian     : 100.0 %
Water Streak       : 1 Hari 🍷
=====
```

Fig. 4. Dashboard menu

If the user has inputted false data, or wishes to restart the daily water intake progress, the user will have the option to reset to a new day through menu 3. Once the user chooses this option, the program will reset the daily progress to 0% and will either reset the water streak if the user hadn’t achieved their daily water intake target or keep the water streak if the user had achieved their daily water intake target.

```
[!] Memulai Hari Baru...
[!] Target kemarin tidak tercapai. Streak di-reset ke 0.
[+] Progres harian di-reset ke 0 ml.
```

Fig. 5. Daily reset option and water streak reset

```
[!] Memulai Hari Baru...
[+] Progres harian di-reset ke 0 ml.
```

Fig. 6. Daily reset option and water streak continues

These tests demonstrate that the program runs successfully as it is able to calculate the user's personalized target, record the user's daily water intake, set a reminder for the remaining daily water intake, and maintain the user's water streak.

IV. ANALYSIS AND DISCUSSION

A. Strengths

Using a single struct keeps the state centralized and avoids global variables. Passing it by reference let the calculator and logger modify the same data, while the const reference in the dashboard function protects data from accidental change. The calculation is transparent and easy to verify by hand, and the output messages give immediate feedback to the user.

B. Sensitivity of the Target

Body weight dominates the result: each additional kilogram adds 35 ml, so a 10 kg difference changes the target by 350 ml, equal to one step of activity. The climate adjustment ranges over only 300 ml, so moving from a subtropical to a tropical climate has the same effect as roughly 8.6 kg of extra body weight. Heavy activity adds up to 700 ml. The resulting targets are typically a few liters per day, the same order of magnitude as total water recommendations in the literature [1], [2], although those values include water from food.

C. Reminder Interval

The default reminder interval is 2 hours. Assuming 16 waking hours, this gives about eight reminders per day. For example, a 3,000 ml target would require roughly 375 ml per reminder. This is a useful guide for pacing, although the program only stores the interval and does not itself run a timer.

D. Limitations

- No input validation: negative or zero weight, non-numeric input, or an invalid menu choice can produce wrong results or leave cin in a failed state.

- The climate and activity adjustments are fixed approximations and do not account for age, sex, pregnancy, health conditions, or fluid from food.
- Progress and streak are held in memory only and are lost when the program closes. The new-day reset (menu 3) is manual, so the program does not detect the date automatically.
- The target check in `logWaterDrink()` uses `>=`, so every drink logged after the target is reached increments the streak again, allowing one day to count several times.
- The program has no real-time clock, so reminders are not triggered automatically.

E. Possible Improvements

Future versions could add input validation loops, save data to a file so the streak persists across days, compare dates to reset progress daily, add a timer or system notification for reminders, and offer more climate and activity categories.

V. CONCLUSION

This paper presented a C++ water-intake application that calculates a personalized daily target using body weight, climate, and activity level, logs drinks, and tracks a daily streak. It also provides a menu with a dashboard and a manual new-day reset. The design is simple, modular, and easy to verify, but it would benefit from input validation, persistent storage, and an automatic reminder mechanism to be practical for daily use.

REFERENCES

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